

Arpan Pathak

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📄 SUMMARY

Systems Software Engineer with **8 years** architecting high-performance, low-latency systems at **Amazon**, **Microsoft**, and **Oracle** – database internals, kernel-level networking, distributed systems, and real-time ML inference. Deep expertise in **Rust**, **C++**, **Go**, and **C**: lock-free concurrency, memory safety, eBPF/Cilium, and GPU-accelerated computing. Shipped measurable wins: **~40% lower P99 latency**, **2x throughput**, **45,000 ops/s** throughput gains, **5,000 QPS** inference serving, and **\$5.4M/month** revenue impact.

🏢 EXPERIENCE

Software Developer II, Microsoft, Redmond, WA

• C++/C# → Rust Migration (Memory Safety):

Spearheaded migration of the Information Protection service from **unsafe C++** and **C#** to **Rust**, moving memory safety to compile time – eliminated use-after-free, buffer-overflow, and null-safety bugs in the migrated path. Delivered **~40% lower P99 latency**, **2x throughput**, **3x lower memory** per request, and **6-figure annual cost savings**.

• Zero-Trust Kernel Networking (eBPF/Cilium):

Architected zero-trust sovereign cloud networking with **eBPF + Cilium** network policies enforcing **data residency** and **SecNumCloud** compliance; authored **Kubernetes operators (Kubebuilder)** to reconcile policy state across AKS clusters in France and Germany.

• AI-Powered Policy Engine:

Built an AI engine that auto-generates Cilium policies and scans for misconfigurations – manual review cut from **4 hours to near-real-time**, zero policy drift across sovereign regions.

• CI/CD & Observability:

Containerized microservices with blue-green deployments, **Prometheus/Azure Monitor**, and automated validation for reliable multi-region releases.

🔗 **TechStack:** **Rust**, **C++**, **eBPF**, **Cilium**, **Kubernetes (AKS)**, **Kubebuilder**, **Prometheus**, **Linux**

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 – Mar 2025

• High-Performance Database Internals:

Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to cut contention in high-concurrency read-write paths – **45,000 ops/s** transaction throughput gain.

• Infrastructure-as-Code SDK:

Engineered a **Terraform provider** (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control-plane tooling with **\$1.2M/month** projected savings.

• Security & Key Management:

Integrated OCI Security Vault across control-plane services for secret management and automated key rotation.

🔗 **TechStack:** **Rust**, **C++**, **Go**, **Terraform**, **PostgreSQL internals**, **OCI**

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 – Oct 2024

• Real-Time ML Inference Engine:

Architected **XGBoost + Deep Learning ensemble** ranking for ads & deals across FreeVe, Twitch, Prime Video, and Amazon Retail – millions of events at **100ms inference latency**, driving **5% CTR lift**, **30%** more deal impressions, and **\$5.4M/month** revenue growth.

• BERT Inference Service:

Delivered **BERT-based** NLP inference handling **5,000 QPS** at peak, cutting support-ticket triage SLA from **7 days to 2 hours**.

🔗 PERSONAL PROJECTS

📷 **Driving-CivicSense (Edge AI Vision)** | 📄 github.com/arpanpathak/driving-civicsense-vision-model | 2026

- On-device AI perception system built in **Rust** for intersection discipline and lane awareness, running entirely on-device (**< 50ms inference latency**, no cloud dependency) for aftermarket dashcam and AR-glasses form factors.
- Fine-tunes **YOLOv8/v11** and custom **CNN** architectures; applies model pruning and quantization to fit tight memory and power budgets on low-power ARM single-board computers (Raspberry Pi Zero, Raspberry Pi 5, and other low-power ARM processor boards).
- Trains on cloud GPU infrastructure with synthetic data augmentation; planning to release a public driving-behavior dataset on **Hugging Face**.

📖 **Companion Book: Seeing Machines: Deep Learning & Computer Vision from Python to Bare Metal** | arpanpathak.github.io/seeing-machines-book

📖 **Physics & Mathematics of Game Development in Rust (Book)** | arpanpathak.github.io/bevy-physics-book | 2026

📖 **CUDA Kernels: GPU & Parallel Programming from First Principles (Book)** | arpanpathak.github.io/gpu-parallel-book | github.com/arpanpathak/gpu-parallel-book | 2025 – Aug 2026

- **Distributed Systems & SDK Design:** Designed uniqueness-constraint indexing and an **SDK** for a multi-tenant NoSQL store using **two-phase commit**, achieving **serializable isolation** at **3K writes/second** – a developer-facing library used by multiple publisher teams.
 - **Event-Driven Ads (CDC):** Delivered programmatic guaranteed ad deals via real-time **change data capture (CDC)** with **< 50ms latency** for auction and programmatic buying platforms.
 - **Big Data Platform:** Built **1-petabyte** data lake & warehouse with **Apache Spark**, **AWS Glue**, Kotlin Spark API, and a **DSL execution engine** for OLTP and BI use cases.
- ⟷ **TechStack:** **Kotlin, Java, Python, PyTorch, XGBoost, Spark, Kafka, DynamoDB, Redshift, AWS**

Software Development Engineer, Razorpay, Bengaluru, India

Aug 2020 – Feb 2021

- Built UPI Payment Gateway Aggregator in **Go** + Protobuf processing **1M+ hourly transactions** with **99.99% uptime** at **150ms P99** under **50,000 TPM** peak load.
- ⟷ **TechStack:** **Go, Redis, PostgreSQL, Protobuf, Kafka**

Software Engineer, Mindfire Solutions, Bhubaneswar, India

Aug 2018 – Feb 2020

- Delivered full-stack workflow tools and an **AR-based product search microservice** serving **10,000+ daily queries**; cut dashboard latency from **5 min to 1 min**.
- ⟷ **TechStack:** **Java, Spring Boot, Angular, MySQL, MongoDB**

SKILLS

⟷ **Languages:** Rust, C++, Go, C, Python, Java, Kotlin, TypeScript, Shell Script

⚙️ **Systems Programming & Performance:** Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

🌐 **Networking & Protocols:** TCP/IP, UDP, TLS, mTLS, DNS, HTTP/2, HTTP/3, QUIC, WebSockets, Socket Programming, Network Performance Optimization

🔒 **Network & Cloud Security:** eBPF, XDP, Cilium, Kernel-Level Networking, Packet Processing, VPC, Subnets, Routing, NAT, VPN, Peering, Load Balancing, Security Groups / Firewalls

📦 **Distributed Systems & Streaming:** Distributed Systems (consensus, replication, sharding), Apache Kafka, Apache Flink, Apache Spark, Real-Time Data Pipelines, Change Data Capture (CDC), Microservices

🔗 **APIs & Serialization:** gRPC, REST API Design, Protobuf

⚡ **ML Inference & AI:** PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), LLMs, ONNX Runtime, On-Device/Edge Inference, Model Pruning & Quantization, Computer Vision (YOLOv8/v11), ANN, CNN, NLP, Deep Learning, Recommendation Systems, MLOps

⚙️ **GPU & Parallel Computing:** CUDA, CUDA-Oxide (Rust-to-CUDA), PTX, SIMT, GPU Kernel Programming, Parallel Computing, GPU-Accelerated Computing

☁️ **Cloud & Infra:** AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS, Bicep), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio, Kubernetes Gateway API

📊 **Observability:** Prometheus, Grafana, Azure Monitor, AWS CloudWatch

📄 **Data & Storage:** DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

🔧 **Core CS:** Data Structures & Algorithms, System Design, Operating Systems, Linux, Computer Architecture

EDUCATION

Bachelor of Technology in Computer Science & Engineering

RCC Institute of Information Technology, Kolkata, India | 2018